Parth Danve

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Education

University of Connecticut — Storrs, CT
B.S. in Computer Science (Honors) — August 2023 – May 2027

- **GPA:** 4.00/4.00, Dean's List all semesters, Babbidge Scholar '24 '25
- **Related Coursework:** OOP in Python, Systems Programming with C, Data Structures, Algorithms, Quantum Computation, Quantum Mechanics, PDE, Software Engineering, Cybersecurity, Machine Learning, Big Data Analytics

Skills

Languages: Python, C++, C, HTML, CSS, Typescript
Python Libraries/SDKs: qiskit, cirq, Gemini API, LangChain, requests, os, sys, Flask, pandas, NumPy, matplotlib, PyTorch, sklearn
Developer Tools: Git, Docker, BurpSuite, Google Cloud Platform, AWS EC2, VS Code, Anaconda, Ollama, Claude Code
Certifications: Qiskit Advocate (IBM), Qiskit Global Summer School 2025: Quantum Excellence (IBM), Quantum Algorithms for PDEs (WISER)
Technical Knowledge: Quantum Optimization, Quantum Inspired Algorithms, Quantum Security, Quantum Networks, Operating Systems, Neural Networks, Machine Learning Models, LLMs, Data Preprocessing, REST APIs

Experience

Quantum Software Engineer Intern — May 2026 – August 2026
IBM — Yorktown Heights, NY

- Developed a Qiskit Serverless function for Trotterization + Approximate Quantum Compilation (AQC), enabling reusable 1-D Hamiltonian dynamics simulations across local, simulated, and IBM Quantum hardware backends.
- Reduced peak memory usage in AQC workloads by $10\times$ at 30–40 qubits and upto $100\times$ at 50 qubits, contributing upstream improvements for scalable quantum simulation.
- Validated the function against an IBM quantum simulation benchmark with $\geq$0.999 AQC fidelity and developed deployment examples and tutorials for the IBM Quantum Platform learning catalog.

Quantum Computing Intern — June 2025 – May 2026
BosonQ Psi (BQP) — Syracuse, NY (Hybrid)

- Implemented Satellite Orbital Propgation mechanisms , via Quantum-Assisted PINNs (QA-PINN) achieving a 200x speedup over Orekit, a leading astrodynamics software while maintaining comparable accuracy.
- Applied Quantum-Inspired Evolutionary Optimization (QIEO) to non-convex sparse recovery problems, reconstructing high-dimensional sparse signals from limited measurements and benchmarking performance against classical optimization baselines.
- Built PyTorch-compatible optimizer wrappers for QIEO, enabling seamless integration with neural network training workflows similar to standard optimizers such as ADAM.

Undergraduate Research Assistant — November 2024 - Present
University of Connecticut - School of Computing — Storrs, CT

- Used qiskit to implement Hybrid Quantum Algorithms asymptotically faster than current heuristics for finding close neighbors in N-Body Simulations. Optimized code to achieve 90% runtime speedup over original implementation (Mentor: Dr. Sanguthevar Rajasekaran)
- Solved Mixed Binary Optimization (MBO) Problems by combining Alternate Direction Method of Multipliers (ADMM) and Quantum Appoximate Optimization Algorithm (QAOA). Benchmarked solutions in qiskit (Mentor: Dr. Bing Wang)

Projects

MIT iQuHack 2025 (3rd Place in D-Wave Challenge) (GitHub) — February 2025
Massachusetts Institute of Technology — Cambridge,MA

- Formulated Quadratic Assignment Problems in a hospital scenario with patient rooms and supply closets having dynamic flow over time as Constrained Quadratic Models and Non-Linear Models to implement Quantum Annealing
- Solved the models using D-Wave's hybrid Quantum Annealers achieving a 50% improvement in the minimization accuracy compared to existing non-linear optimization methods in Scipy.

Yale Quantum Hackthon 2025 (1st Place in BlueQubit Challenge) (GitHub) — March 2025
Yale University — New Haven, CT

- Worked with peaked quantum circuit sampling. Challenge was to find the hidden peaked bitstring in circuits that were very hard to run on classical CPU/GPU simulators and some of them even on actual quantum hardware due to extensive entanglement and depth
- Utilized BlueQubit, Qiskit-Aer and Quimb Libraries, to experiment with Matrix Product State (MPS) Simulations by tuning bond dimensions, acheiving a 500% speedup on a 44 qubit circuit.

Quantum Advisors (1st Place HackUConn 2025) (GitHub) — March 2025
University of Connecticut — Storrs, CT

- Optimized university finals scheduling using the constrained quadratic model in D-Wave's Quantum Annealers. Simulated annealing through variational quantum algorithms in IBM-Qiskit. Finetuned GPT-2 to prototype the concept of AI assisted University Academic Advising